

Sample

Instructions

- Quiz 1 is a paper test, taken offline and answered on a Scantron sheet. No laptops, phones or other devices.
- Bring a pencil. Scantron sheets are read by machine and need pencil, not pen.
- You may use one crib sheet: a single 8.5 × 11 inch page, filled on both sides if you like. Nothing else.
- Each question has one best answer, A to E. Fill in one bubble per question on the Scantron sheet; only the Scantron sheet is graded.

1. In the Homework 1 stack, web runs Apache with PHP and db runs MySQL. A browser asks for a page that lists rows from a table. In which order is the request handled?

- A. Browser → the PHP code, which runs in the browser and asks MySQL directly
- B. Browser → Apache → browser; the browser then asks MySQL for the rows
- C. Browser → MySQL → the PHP code → Apache → browser
- D. Browser → Apache → MySQL → browser; the PHP code runs afterwards, in the browser
- E. **Browser → Apache → the PHP code → MySQL → the PHP code builds the HTML → Apache → browser**

The browser only ever speaks HTTP to the web server. PHP runs on the server, and only it talks to the database.

2. "The image you tested is the image that ships." Which problem does that most directly address?

- A. Forms that accept invalid input
- B. CSS rules that apply to every page
- C. Pages that load slowly on phones
- D. **Environment drift: laptop, staging and production quietly becoming different**
- E. Passwords committed to a public repository

3. What is continuous integration?

- A. Servers are described in files and reviewed like code
- B. Containers are restarted when they crash
- C. Code is deployed automatically whenever the tests pass
- D. Every developer works on the same branch at the same time
- E. **Every push is built and tested automatically**

4. As the course defined it, a framework is...

- A. a container image with a web server already installed
- B. a tool that packages an application with its operating-system dependencies
- C. a library of functions you call when you need them
- D. **the repeated plumbing of web applications, written once, with opinions**
- E. a programming language designed for the web

5. A service has ports: ["3000:80"]. Which is true?

- A. The container listens on ports 80 through 3000
- B. Both the laptop and the container use port 3000
- C. **Your laptop's browser uses http://localhost:3000; the server inside the container is listening on port 80**
- D. Port 80 on your laptop is forwarded to port 3000 in the container
- E. Your laptop's browser uses http://localhost:80; the container listens on 3000

Host port on the left, container port on the right.

6. A container is best described as...

- A. a lightweight virtual machine with its own guest operating system and kernel
- B. a compressed archive of an application's source code
- C. **one or more processes isolated with namespaces and cgroups, sharing the host's kernel**
- D. a server rented from a cloud provider by the hour
- E. a copy of the host's hard disk, taken when it starts

7. Which is the recommended rule of thumb for mounts?

- A. Both in bind mounts, so that they are in git
- B. Database files in a bind mount; source code in a named volume
- C. Neither: copy everything into the image and rebuild after each edit
- D. Both in named volumes, so that Docker manages them
- E. **Source code in a bind mount; database files in a named volume**

8. Someone deletes the assert line from this test, leaving the other two lines. Later, a bug makes add() subtract. What does PHPUnit report for the test?

```
public function test_buying_adds_to_the_amount(): void
{
    $stock = new Stock(2);
    $stock->add(3);
    $this->assertSame(5.0, $stock->amount());
}
```

- A. **No failure. It only flags the test as risky: "This test did not perform any assertions"**
- B. A failure, because the amount is wrong
- C. The test is skipped, and so are all the tests after it
- D. An error: every test must contain an assertion, so the file does not load
- E. Failed asserting that -1.0 is identical to 5.0.

9. A Playwright test creates its product with the name Apples `${Date.now()}`. Why include `Date.now()`?

- A. So the products are listed in the order they were created
- B. **So the test's data can't collide with data left by other tests or earlier runs**
- C. To slow the test down so the page can load
- D. To check that the application formats dates correctly
- E. Because Playwright requires every string to be unique

Every test starts from nothing, so each makes its own data.

10. In this PHPUnit test, which line is the "act" step of arrange, act, assert?

```
public function test_buying_adds_to_the_amount(): void
{
    $stock = new Stock(2);
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    $this->assertSame(5.0, $stock->amount());
}
```

- A. **`$stock->add(3);`**
- B. `$stock = new Stock(2);`
- C. `$this->assertSame(5.0, $stock->amount());`
- D. `public function test_buying_adds_to_the_amount(): void`
- E. All three lines together

11. In docker `compose run --rm unit`, what does `--rm` do?

- A. Runs the tests in a random order
- B. **Deletes the one-off container when the run finishes**
- C. Removes the project's volumes
- D. Removes the image after the tests pass
- E. Re-runs any test that failed

12. Tailwind's md breakpoint is 768px. At which window widths can you see this element?

```
<aside class="hidden md:block">Filters</aside>
```

- A. **768px and wider**
- B. Narrower than 768px
- C. Only between 768px and 1024px
- D. At no width: hidden overrides md:block
- E. At every width: md:block overrides hidden

hidden applies everywhere; md:block takes over from 768px up. Mobile first.

13. What does a Tailwind class name such as rounded-full or px-2 describe?

- A. What the element is
- B. Which breakpoint the element belongs to
- C. Which JavaScript component owns the element
- D. **What the element looks like**
- E. Which page the element is on

14. In this route, who calls the function that returns the greeting, and when?

```
Route::get('/hello', function () {  
    return 'Hello, Laravel!';  
});
```

- A. PHP, once, when the file is first loaded
- B. **Laravel's router, when a GET request for /hello arrives**
- C. The browser, which downloads the function and runs it
- D. Nobody: the function is only there as documentation
- E. Your own code, on the line after the route is defined

Inversion of control: the framework owns the flow and calls the handler you registered.

15. A visitor saved this as a note: `<b>hi</b>`. A Blade template prints it with `{{ $note }}`. What does the page show?

- A. **The characters `<b>hi</b>`, as visible text**
- B. An error page: the note is rejected
- C. Nothing: Blade removes anything that contains a tag
- D. The word hi, in bold
- E. The word hi, with no tags and no bold

Escaping turns `<` and `>` into entities, so the browser displays them instead of treating them as HTML.